Phase 4 Delivery

DAEN 460 Capstone Senior Design

Alexander Abuabara

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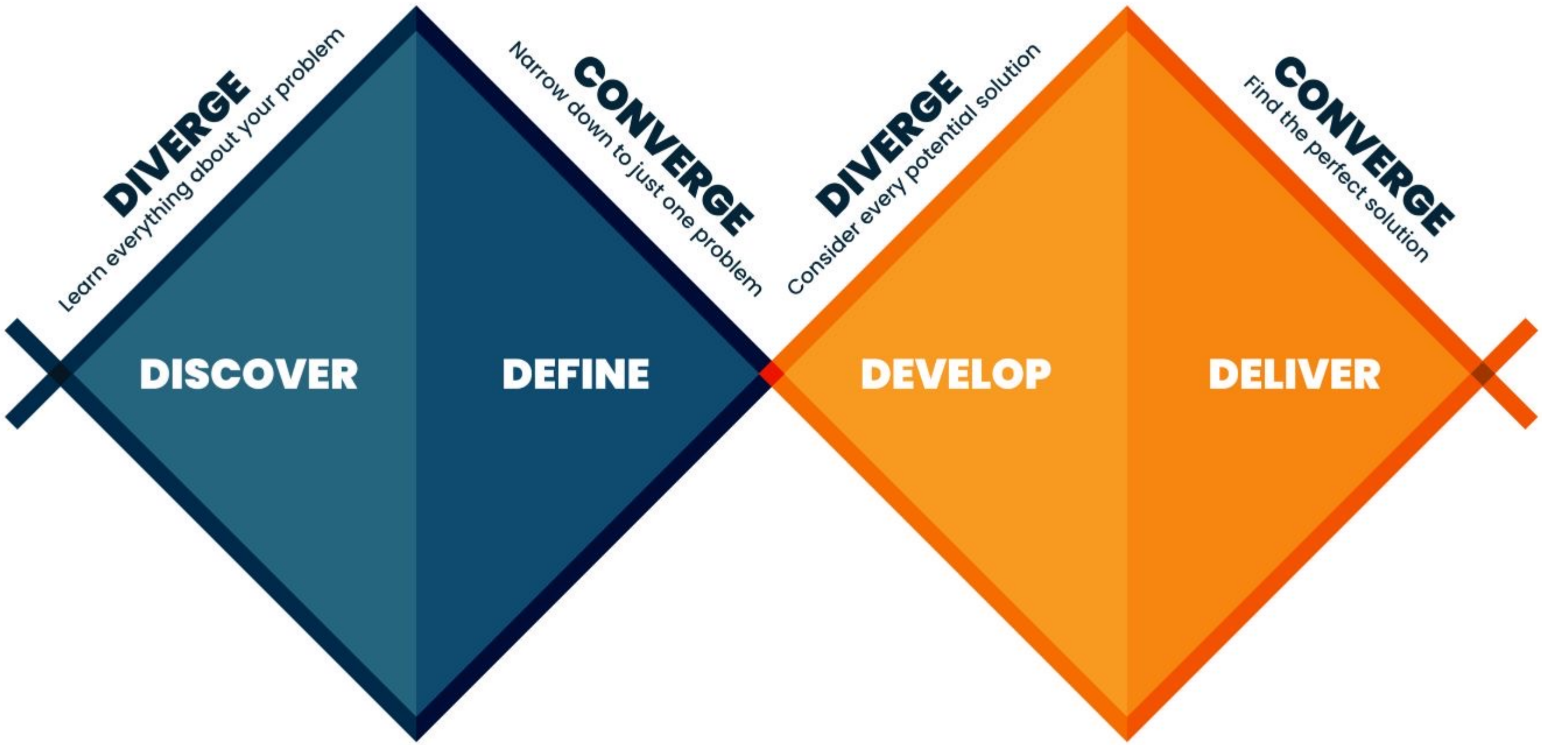
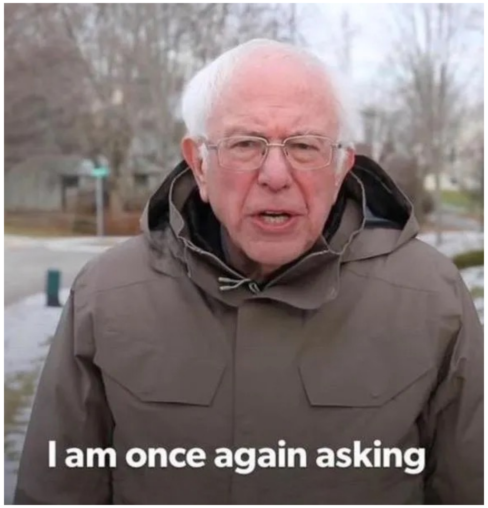
ENGINEERING
TEXAS A&M UNIVERSITY


DIVERGE
Learn everything about your problem
CONVERGE
Narrow down to just one problem
DIVERGE
Consider every potential solution
CONVERGE
Find the perfect solution
DISCOVER
DEFINE
DEVELOP
DELIVER

Today's Agenda

- **Exit Interviews** **(everyone needs to be here next week!)**
  ISEN UG Program Staff
- **Comms activity**
  - Final Report Template and Expectations
  - Ethics
- **Phase 4 Process and Requirements**
  - Revise
    - **Constrains**
    - **Standards**



- **Critique** your own project against **learning outcomes** rather than technical checklists only!
  - You are **not** being graded on **reproducing** a solution but on the **quality of your thinking and decisions**!
- You're not just working alongside your peers, you're learning with them.
  - The goal is to move from individual results to shared success through real collaboration.
  - Peers are teammates, not just people doing the same assignment.
  - The aim is to build a culture of shared responsibility and genuine collaboration.
- Pay attention to **how** and **why** you think the way you do when learning!

Learning Outcomes *(syllabus)*
Ability to:

1. **Identify**, **formulate**, and **solve** complex engineering problems by applying principles of engineering, science, and mathematics.
2. **Apply engineering design** to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. **Communicate effectively** with a range of audiences.
4. **Recognize ethical and professional responsibilities** in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. **Function effectively on a team** whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks and meet objectives.
6. **Develop and conduct appropriate experimentation, analyze and interpret data** and use engineering judgment to **draw conclusions**.
7. **Acquire and apply new knowledge as needed**, using appropriate learning strategies.

Upcoming Semester Schedule

- 4/14 **Phase 4 Presentations BH/COE**
- 4/14 mandatory attendance for all 2:15-3:15 **Senior Exit Interviews**
- 4/21 **Phase 4 Presentations ML**
- 4/24 in-person, all day **COE Showcase**
  schedule at https://tamueps.secure-platform.com/

8:00 AM – 10:00 AM	10:30 AM – 1:30 PM	1:30 PM – 3:00 PM	3:00 PM
Team Registration/Booth Setup, *ConocoPhillips Atrium, ZACH*	Present to assigned judges, *ZACH*	Interact with industry guests, *ZACH*	Awards Ceremony, *Chevron Rooms, ZACH*

- 5/5 5 PM **Final Report Due**

Final/Phase 4 Presentation

- Welcome to invite sponsors to attend your presentation
- Coordinate if you need to arrange parking for them

Peer Evaluation

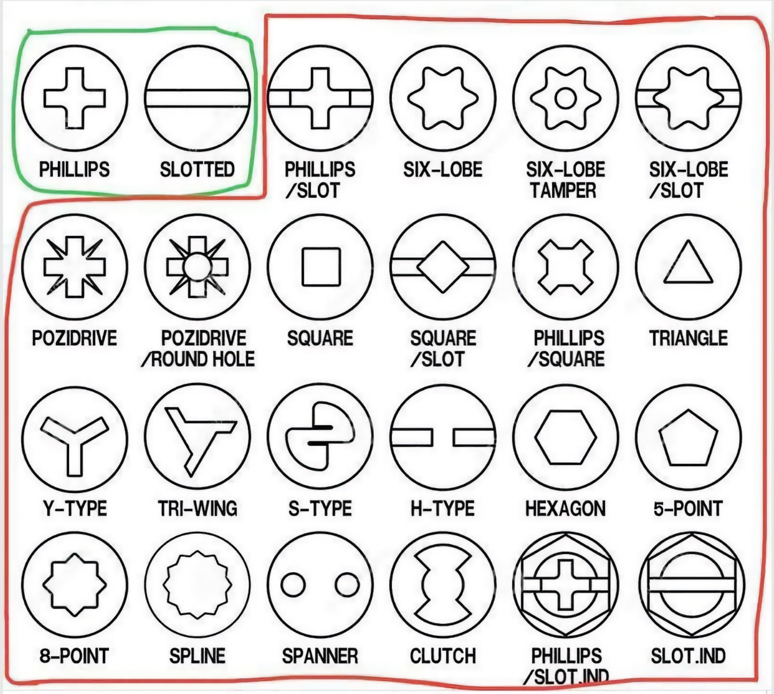
- **Forms**
  - Peer Evaluation due after the final report
  - You will rate your performance on the project as well as teammates
- **A grade (12% of overall grade) determined using:**
  - Peer Evaluation ratings and comments
  - Class attendance/tardiness/participation
  - Instructors' observation of team participation/contribution

Comms Activity

IT NEEDS TO BE REVIEWED IN ALL PROJECTS!

Standards (A Universal Language)

- Standards are the agreed-upon rules and technical specifications that ensure safety, reliability, and interoperability. Without them, your lightbulb might not fit into a socket made by a different company.
- **Safety & Compliance**: Regulatory bodies (like OSHA or the FAA) mandate standards to prevent catastrophic failure and protect the public.
- **Interchangeability**: Standards like ISO or ASME ensure that parts from different global suppliers work together seamlessly.
- **Quality Benchmarks**: Following established standards provides a baseline for "what good looks like," reducing the risk of design flaws.



Screw standards define the dimensions, thread pitch, and tolerances for fasteners to ensure interchangeability and strength, with the two primary systems being the Unified Thread Standard (UTS/Imperial) in North America and ISO Metric worldwide. Key standards include ISO 68-1 for metric, and ASME B1.1 for unified inch threads (UNC/UNF).

IT NEEDS TO BE REVIEWED IN ALL PROJECTS!

Engineering Standards

- **Data Quality & Governance**
  - ISO/IEC 8000 – **Data Quality Standard**: Defines requirements for reliable and accurate data across systems.
  - **ISO/IEC 11179 – Metadata Registry Standard**: Governs how data elements and metadata should be defined and documented.
  - **ISO/IEC 38505 – Data Governance Framework**: Defines governance practices for managing organizational data.
  - **General Data Protection Regulation** (GDPR)
- **Data Management Body of Knowledge**
  - DAMA International Data Management Framework (DMBOK): Widely used reference covering: data governance, data architecture, data quality, master data management, metadata management
- **Data Architecture**
  - The Open Group TOGAF Enterprise architecture framework often used in data platform design.
  - World Wide Web Consortium standards (JSON, XML, RDF, etc.) that govern structured data exchange.
  - ISO/IEC 9075 – SQL Standard: The formal specification behind relational databases.
- **Data Security Standards**
  - **ISO/IEC 27001 – Information Security Management**
  - National Institute of Standards and Technology Cybersecurity Framework

  These can govern encryption, access control, auditing, breach management.

IT NEEDS TO BE REVIEWED IN ALL PROJECTS!

Engineering Standards

- **Manufacturing**: ASME Y14.5 (GD&T), ASTM Standards, ISO 2768 (General Tolerances), ISO 9001, ANSI Standards, IPC Standards, API Standards
- **Aerospace & Defense**: AS9100, ASME Standards, ASTM F519, MIL-STD-810 (Environmental Engineering), MIL-STD-461 (EMC), RTCA DO-160 (Environmental Testing), SAE AS Standards, ISO 9100 Series
- **Healthcare**: ISO 13485, IEC 60601 (Medical Electrical Equipment), ISO 14971 (Risk Management), ASTM Standards (Medical Devices), FDA 21 CFR Part 820 (Quality Systems Regulation), ANSI/AAMI Standards
- **Food Industry**: ISO 22000 (Food Safety Management), 3-A Sanitary Standards, ANSI/NSF Standards, ASTM Food Packaging Standards, EHEDG (European Hygienic Engineering & Design Group), GMP Engineering Standards

Phase 4

- Phase 4 is about delivering the results of your project
- You should provide **actionable information** that will allow your <u>sponsor to implement</u> your solution <u>and/or</u> <u>continue development</u> based on your research
- Your presentation should provide **enough analysis** to prove you have rigorously examined the problem and your solution
- You should also show that you have considered **appropriate engineering standards and constraints** as you developed your solution
- Phase 4 technical grade will **not be subject to resubmissions**

Phase 4 Guidelines

On Canvas

Phase 4 Presentation Guidelines

Project Recap

- In addition to displaying your problem statement in the introduction, provide a project recap
- A slide or two (visuals are good) explaining the "why" of your project and your efforts to date
- This is likely going to be included in your showcase poster as a project introduction
- Think of it as your "elevator pitch", but along with a minute or so to talk, you'll have visuals to back it up

Phase 4 Presentation Guidelines

Final Design Architecture

- Explain your solution, focus on functionality, keep technical details to a minimum
- Map your requirements to your solution, indicate how each requirement is satisfied by the design (focus on key requirements if you have many)
- If your results are research focused, identify areas of research and map to key requirements

Phase 4 Presentation Guidelines

Analysis

- Analyze your final solution using appropriate criteria (some may be same used in Phase 3)
- Economic, performance, risk are three areas
  - Define each clearly as they are presented
- Other areas of analysis should be explored as appropriate for your project
- For research, discuss the impact of the research and analysis of validity

Phase 4 Presentation Guidelines

Engineering Standards and Constraints

- Describe any engineering standards that guided your solution and discuss relevance
- Explain any constraints faced by your team/project and how they are addressed
- See guidelines document for tips

Phase 4 Presentation Guidelines

Conclusion and Recommendations

- Finalize the presentation by summarizing what your outcomes indicate (don't go through final design again, but reference earlier information as needed to "wrap up" your results)
- Include any implementation plan you have created
- If future work to continue/expand your project is likely, indicate areas of focus that might take

Showcase Tips

- Do very **brief introductions**
- **Tell a story**, not get bogged down in technical details
- **Focus on the process** you went through, not the details of each step
- State the **requirements** of your solution clearly
- Identify **multiple solutions** you may have explored, but don't spend a lot of time detailing them
- **Explain your solution** and go through the analysis of how it meets your requirements
- **End with a strong conclusion**, if there are relevant metrics proving the validity of your solution, present them!

Showcase Tips

- **Check requirements**
  - All team members should be present (but not at the same time)
  - Limit presenters to 2-3 people is fine if desired
- **Judging criteria**
  - Problem definition (25%)
  - Design methodology (25%)
  - Achieved or expected outcomes (25%)
  - Overall quality of the presentation (25%)

Future

The people who are crazy enough to think they can change the world are the ones who do.

Steve Jobs "Think Different" Campaign Speech 09/1997
https://youtube.com/watch?v=b4n8uT12ij8&t=410